

ServiceNow Incident Lifecycle Automation

1. Ideation Phase

1.1 Project Title

Incident Lifecycle Automation in ServiceNow

1.2 Project Overview

The project is mainly focused on automating and standardizing the Incident Management lifecycle in ServiceNow. It helps Service Desk agents to create, classify, assign, investigate, resolve, and monitor incidents in an organized and systematic way.

The solution also combines Knowledge Management, Change Management, SLA tracking, and Child Incident Management to help resolve incidents faster and ensure that all incidents are properly tracked throughout the lifecycle.

1.3 Problem Statement

In many organizations, IT issues such as network failures, VPN problems, hardware issues, and application errors are handled through emails, calls, or informal communication. This can lead to:

- Improper incident classification.
- Delayed assignment to appropriate support teams.
- Difficulty in finding existing solutions.
- Limited visibility into incident status. **Project Planning Phase**
- Delayed escalation and resolution.
- Difficulty tracking related changes and incidents.
- Incomplete cause and resolution details.
- SLA compliance issues.

Therefore, a centralized and standardized incident management process is needed.

1.4 Proposed Idea

The proposed solution uses ServiceNow to manage the incident lifecycle through the following process:

Incident Creation → Classification → Knowledge Assistance → Assignment → Investigation → Change/Resolution → Knowledge Creation → Closure

The system supports incident creation, classification, Knowledge Article recommendations, reassignment to Level 2 teams, Emergency Change integration, Child Incident creation, resolution documentation, Knowledge Article creation, and SLA tracking.

1.5 Project Goals

1. The main goals of the project are:
2. Standardize incident creation and classification.
3. Improve incident assignment and escalation.
4. Reduce resolution time using Knowledge Management.
5. Integrate Incident and Change Management.
6. Track related issues using Child Incidents.
7. Maintain proper cause and resolution details.
8. Improve SLA monitoring and service visibility.
9. Provide a complete and traceable incident lifecycle.

Phase 2: Requirement Analysis

2.1 Business Requirements

Requirement	Description
Incident Creation	Service Desk agents should be able to create incidents quickly and in a consistent way.
Incident Classification	Each incident should include details such as category, subcategory, urgency, service, and configuration item to make sure it is routed correctly from the beginning.
Knowledge Management Integration	Agents should be able to access and attach relevant knowledge articles so that incidents can be resolved faster.
Assignment & Reassignment	Incidents should be routable to the correct support group, with the ability to reassign to Level 2 teams when needed.
Escalation & Investigation	Level 2 technicians should have complete access to the incident details, history, and SLA status so they can investigate the issue and resolve it effectively.
Change Integration	Incidents requiring a technical fix should support direct creation and linking of emergency change requests.
Child Incident Tracking	Recurring or related issues should be tracked using child incidents that are linked to a parent ticket.
Resolution Documentation	Cause and resolution details should be captured before an incident is closed.
Knowledge Article Generation	Resolved incidents should be convertible into knowledge articles for future reuse.
SLA & Compliance Visibility	SLA status should be visible throughout the entire incident lifecycle to help with compliance tracking and reporting.

2.2 Functional Requirements

FR-01: Service Creation

- **Service:** Remote Access
- **Operational Status:** Operational

FR-02: Service Offering Creation

- **Service Offering:** Corporate VPN
- **Parent Service:** Remote Access
- **Operational Status:** Operational

FR-03: Incident Creation

Service Desk agents should be able to create an incident capturing:

- Short Description
- Caller
- Assignment Group
- Description
- Channel
- Category
- Subcategory
- Urgency
- Service
- Service Offering
- Configuration Item

FR-04: Incident Classification

Incidents should be classified with accurate values:

- **Category:** Network
- **Subcategory:** VPN
- **Urgency:** 2 – Medium
- **Service:** Remote Access
- **Service Offering:** Corporate VPN
- **Configuration Item:** an assigned workstation asset

FR-05: Knowledge Integration

Agent Assist should surface relevant knowledge articles based on the incident description. Agents should be able to:

- View recommended articles
- Open an article
- Mark an article as Helpful
- Attach the article to the incident
- Add comments regarding the article

FR-06: Watch List and Work Notes

The system should allow stakeholders to be added to:

- Watch List
- Work Notes List

This keeps relevant people informed without transferring ownership of the ticket.

FR-07: Reassignment

Incidents should be re-assignable from Service Desk to the appropriate Level 2 group

Example: **Service Desk** → **Network**

FR-08: Investigation

Level 2 support should be able to:

- View full incident details

- Review knowledge articles
- Review activity history
- View SLA information
- Take ownership of the incident

FR-09: Change Integration

When resolving an incident requires a technical change, the system should support creating and linking an emergency change request. The incident should be placeable in:

- **State:** On Hold
- **Reason:** Awaiting Change

FR-10: Child Incident

The system should allow creation of child incidents for recurring or related issues, linked back to the parent incident.

FR-11: Cause Documentation

The probable cause of the incident should be documented

Example: a suspended service that required a restart.

FR-12: Resolution Documentation

The system should record:

- **Resolution Code:** Workaround provided)
- **Resolution Notes:** service restarted per the linked change request)

FR-13: Knowledge Article Creation

After resolution, the incident should support generating a knowledge article:

- **Knowledge Base:** IT
- **Template:** Standard
- Article should reflect the incident's resolution details

FR-14: SLA Tracking

SLA information should remain visible throughout the incident lifecycle via related records, supporting ongoing compliance tracking.

3. Project Design Phase

3.1 Solution Architecture

The project is developed using ServiceNow's integrated ITSM features to manage the complete incident lifecycle. The main components used in the project are **Incident Management, Knowledge Management, Change Management, Configuration Management, SLA Management, and Child Incident Management.**

High-Level Architecture:



3.2 Service Structure

The service structure used in the project is:

Remote Access



Corporate VPN

- **Remote Access** – Parent Service
- **Corporate VPN** – Service Offering

This service structure gives the incident the required service-related context. ServiceNow Incident Management allows incidents to be associated with services, service offerings, and configuration items.

3.3 Incident Lifecycle Design

The incident is designed to move through a structured lifecycle:

New → In Progress → On Hold → Resolved → Closed

This service structure provides the necessary service details for an incident. ServiceNow Incident Management allows incidents to be linked with services, service offerings, and configuration items.

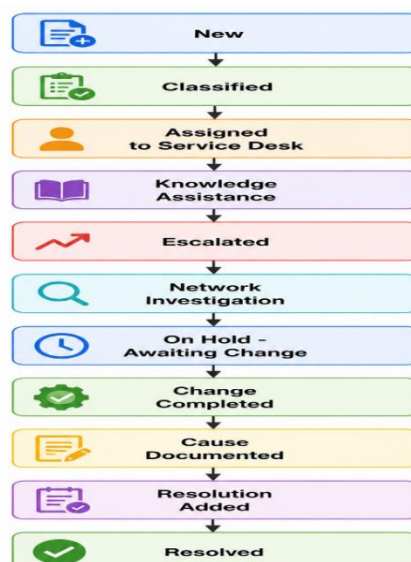


Figure 3.2: Incident Lifecycle Design 1

3.4 Stakeholder Design

Stakeholder	Role	Expected Benefit
End User	Reports IT issues	Faster resolution and better visibility
Service Desk Agent	Creates and classifies incidents	Standardized incident handling
Level 2 Support	Investigates and resolves incidents	Better diagnosis and ownership
Change Team	Handles required emergency changes	Controlled change implementation
ServiceNow Administrator	Maintains the platform	Consistent and manageable process

3.5 Module Integration

The project integrates the following ServiceNow capabilities:

- **Incident Management** – Incident creation and lifecycle management
- **Knowledge Management** – Knowledge-based troubleshooting and article creation
- **Change Management** – Emergency Change integration
- **Configuration Management** – Service and CI information
- **SLA Management** – SLA monitoring and tracking
- **Related Records** – Changes, Child Incidents, SLAs, and Knowledge Articles

This integration helps maintain a connected and traceable incident management process. ServiceNow also allows a change request to be created directly from an incident whenever a change to an IT service is needed.

4. Project Planning Phase

4.1 Project Execution Roadmap

The project is planned to automate and standardize the complete Incident Management lifecycle in ServiceNow. The project activities are divided into different milestones covering service configuration, incident creation, classification, knowledge assistance, escalation, change management, resolution, and knowledge creation.

Milestone	Planned Activity
Milestone 1	Create Service
Milestone 2	Create Service Offering
Milestone 3	Create Incident
Milestone 4	Classify Incident
Milestone 5	Integrate Knowledge
Milestone 6	Update Watch List
Milestone 7	Reassign Incident
Milestone 8	Level 2 Investigation
Milestone 9	Create Emergency Change
Milestone 10	Create Child Incident
Milestone 11	Add Cause
Milestone 12	Add Resolution
Milestone 13	Create Knowledge Article
Milestone 14	Final Validation
Milestone 15	Testing and Deployment

4.2 Task Planning

Phase A – Service Configuration

Objective: Configure the required service and service offering for the incident.

1. Create **Remote Access** service.
2. Set the operational status.
3. Create **Corporate VPN** service offering.
4. Associate the offering with **Remote Access**.

Deliverable: Remote Access service with Corporate VPN service offering.

Phase B – Incident Management

Objective: Create and classify a VPN-related incident.

1. Impersonate **Service Desk** user.
2. Create a **VPN incident**.

3. Add caller and short description.
4. Assign the incident to **Service Desk**.
5. Update incident details.
6. Select **Network** → **VPN** category.
7. Set urgency and priority.
8. Associate service, service offering, and CI.

Deliverable: Properly created and classified VPN incident.

Phase C – Knowledge Management

Objective: Use existing knowledge to assist in resolving the incident.

1. Open **Agent Assist**.
2. Search for relevant VPN Knowledge Articles.
3. Open the appropriate article.
4. Mark the article as **Helpful**.
5. Attach the article to the incident.
6. Add relevant comments.

Deliverable: Relevant Knowledge Article attached to the incident.

Phase D – Collaboration and Escalation

Objective: Collaborate with stakeholders and escalate the incident to the appropriate support team.

1. Add stakeholder to the **Watch List**.
2. Add user to the **Work Notes List**.
3. Reassign the incident to **Network Support**.
4. Verify the **Assigned To** field.
5. Impersonate **Level 2 Technician**.
6. Assign the incident to the technician.
7. Verify incident and SLA information.

Deliverable: Incident successfully escalated to Level 2 support.

Phase E – Change and Child Incident Management

Objective: Manage additional work required to resolve the incident.

1. Update the **Configuration Item**.
2. Set the incident to **On Hold**.
3. Select **Awaiting Change**.
4. Create and link an **Emergency Change**.
5. Create a **Child Incident**.
6. Verify the parent-child relationship.

Deliverable: Emergency change and child incident linked to the parent incident.

Phase F – Resolution

Objective: Document the cause and resolution and complete the incident.

1. Add the **probable cause**.
2. Add the **resolution code**.
3. Add **resolution notes**.
4. Resolve the parent incident.
5. Verify the child incident status.
6. Verify the change request relationship.

Deliverable: Incident resolved with complete resolution information.

Phase G – Knowledge Creation

Objective: Create reusable knowledge based on the incident resolution.

1. Select **Create Knowledge**.
2. Select **IT Knowledge Base**.
3. Select **Standard** template.
4. Verify the article information.
5. Save the Knowledge Article.
6. Verify the article under **Related Records**.

Deliverable: New Knowledge Article created from the incident resolution.

4.3 Project Workflow



4.4 Project Deliverables

Phase	Deliverable
Service Configuration	Remote Access Service and Corporate VPN Offering
Incident Management	Created and classified VPN Incident

Phase	Deliverable
Knowledge Management	Knowledge Article attached to Incident
Collaboration & Escalation	Incident assigned to Level 2 Network Support
Change Management	Emergency Change linked to Incident
Child Incident Management	Parent-Child Incident relationship
Resolution	Cause and Resolution details
Knowledge Creation	New IT Knowledge Article
Final Phase	Validated and tested project

4.5 Expected Project Outcome

By the end of the project, the Incident Lifecycle Automation in ServiceNow will demonstrate the complete incident management process, starting from service configuration and incident creation to classification, knowledge assistance, escalation, change management, resolution, and knowledge creation. This helps ensure that incidents are managed in a structured and traceable way.

5. Project Development Phase

This phase includes the actual implementation work carried out in ServiceNow.

5.1 Service Configuration

Step 1: Create Service

Navigate to:

All → cmdb_ci_service.list → New

Enter:

Field	Value
Name	Remote Access
Operational Status	Operational

Submit the record.

Service Remote Access

Open in CMDB Workspace

Update

Convert to Service Instance

Delete

NameRemote Access

Model ID

Owned by

Business criticality2 - somewhat critical

Version

Used forProduction

Operational statusOperational

Service classificationBusiness Service

Comments

Approval group

Support group

Change Group

Managed by

SLA

Location

Vendor

Related Items

Search for CI

Open in CMDB Workspace

Update

Convert to Service Instance

Delete

Related Links

Convert to business service

Convert to technology management service

Subscribe

Step 2: Create Service Offering

Navigate to:

All → service_offering.list → New

Enter:

Field	Value
Name	Corporate VPN
Parent	Remote Access
Operational Status	Operational

Submit the record.

Offering Corporate VPN

Name: Corporate VPN

Model ID: [Search]

Owned by: [Search]

* Parent: Remote Access [Search]

Used for: Production

Operational status: Operational

Vendor: [Search]

Approval group: [Search]

Support group: [Search]

Change group: [Search]

Managed by: [Search]

Version: [Search]

Business criticality: 2 - somewhat critical

SLA: [Search]

Location: [Search]

Short Description: [Text Area]

Description: [Text Area]

Comments: [Text Area]

Related Items: [Icons] Show 3 Levels [Search]

Contract: [Search]

Open in CMDB Workspace Update Delete

Related Links
Subscribe

5.2 Incident Creation

Navigate to:

Service Operations Workspace → List → Incidents → Open → New

Create an incident with:

Field Value

Short Description Unable to connect to Corporate VPN from home office

Caller Michael Hoefer

Assignment Group Service Desk

Save the incident.

servicenow All Favorites History Workspaces Service Operations Workspace Search

Create New Incident

Details

Incident

Short description *

Unable to connect to Corporate VPN from home office

Description

Number

INCO010010

State

New

Caller *

Michael Hoefer

Location

SHS quadra 5, Bloco E., Brasilia

Channel

-- None --

Category

Inquiry / Help

Impact

3 - Low

Urgency

3 - Low

Priority

4 - Low

Agent Assist

Unable to connect to Corpor...

Article

How to configure VPN for Apple D...

How to configure VPN for Apple Devices For an iPhone or iPad running iOS? Select Settings...

updated 12 years ago · 3 views

Article

Automatic Replies (Out Of Office)

Automatic Replies (Out Of Office) This article explains how to use automatic replies in Outlo...

updated 4 years ago · 85 views

Article

Excel Functionality

Excel Functionality Copying formulas without changing cell locations When you want to kno...

updated 12 years ago · 2 views

5.3 Incident Classification

Update the incident with:

Field	Value
Description	Michael mentioned that he was able to connect yesterday evening, but this morning he received an authentication error. The Internet connection is working properly.
Channel	Phone
Category	Network
Subcategory	VPN
Urgency	2 – Medium
Service	Remote Access
Service Offering	Corporate VPN
Configuration Item	ThinkStationS20

Add the following work note:

Classified and assigned incident.

Then select **Assign to me** and save.

The screenshot displays the ServiceNow incident management interface. The incident title is "Unable to connect to Corporate VPN from home office" with ID INC0010003. The incident is classified as "Remote Access" under the "Corporate VPN" service offering. The configuration item is "PowerEdge T110 II". The business impact is currently empty. The assignment group is "Network" and it is assigned to "David Loo". The incident state is "Resolved was On Hold". The activity log shows three entries by David Loo: field changes at 10:45:37, 10:45:25, and 10:32:44. The resolution notes indicate a workaround was provided and the VPN service was restarted. The right-hand panel shows record information, including SLAs (Response SLA: Completed, Resolution SLA: Paused) and caller information (Michael Hoefer, Development, Brasilia - 10:23:40, America/Los_Angeles).

5.4 Knowledge Integration (Agent Assist)

Open **Agent Assist** from the right-side panel.

Verify that the appropriate Knowledge Articles are shown based on the incident's short description.

Perform the following:

1. Open a relevant Knowledge Article.
2. Select the three-dot menu.
3. Mark the article as **Helpful**.
4. Select **Attach Article**.

5. Add the comment:

Follow the steps and resolve the incident.

6. Attach the article.

7. Return to the incident.

8. Verify that the article is recorded in the Activity log.

The screenshot shows the ServiceNow interface for an incident titled "Unable to connect to Corporate VPN from home office". The incident is in the "Resolved" state. The description states: "Michael advised he was able to connect yesterday evening, but this morning received an authentication error. The Internet works fine." The incident details include: Number INC0010003, State Resolved, Caller Michael Hoefer, Location SHS quadra 5, Bloco E., Brasília, Channel Phone, Priority 4 - Low, and Category Network. The Watch list includes Samantha Bordwell. The Activity log shows three entries by David Loo: "Field changes" at 10:45:37 (Incident state Resolved was On Hold), "Field changes" at 10:45:25 (Resolution co... Workaround provided was Empty, Resolution no... Restarted VPN-SRV-02 service as per emergency change request was Empty), and "Field changes" at 10:32:44 (Incident state On Hold was In Progress). The Record Information panel shows SLAs and timings (Response SLA Completed, Resolution SLA Paused) and Caller information (Michael Hoefer, Development, Brasília - 10:23:40, America/Los_Angeles).

5.5 Watch List & Work Notes

In the incident **Details** tab:

- Add **Samantha Bordwell** to Watch List.
- Add **Beth Anglin** to Work Notes List.

The screenshot shows the ServiceNow interface for the same incident, but with updates. The Activity log now includes entries for Beth Anglin: "Assigned to David Loo was Empty" at 10:06:52, "Assigned to Empty was Beth Anglin" at 10:06:52, "Comments" at 09:46:06 (Follow the Steps and Resolve the Incident KB0000008 : How to configure VPN for Apple Devices), "Field changes" at 09:31:52 (Assigned to Beth Anglin was Empty, Configuration... ThinkStation S20 was Empty, Incident state In Progress was New), and "Work notes" at 09:31:52 (Classified and assigned incident.). The Watch list still shows Samantha Bordwell. The Record Information panel remains the same.

5.6 Incident Reassignment

Update:

Assignment Group → Network

Verify that the **Assigned To** field is automatically cleared when the assignment group is changed.

Save the incident.

5.7 Level 2 Investigation

Impersonate the Level 2 Network technician.

Navigate to:

Service Operations Workspace → Incidents → Open

Open the VPN incident.

Select **Assign to me**.

Verify:

- Incident description is available.
- Knowledge Articles are available.
- SLA information is visible.
- Task SLAs appear in Related Records.
- Incident information is accessible to the Network team.

5.8 Emergency Change Integration

Update the Configuration Item as required.

Place the incident in:

State: On Hold

On Hold Reason: Awaiting Change

Save

The screenshot displays the ServiceNow interface for a Configuration Item (CI) named 'PowerEdge T110 II'. The interface is divided into several sections:

- Header:** Shows the 'Service Operations Workspace' and a search bar.
- Details Tab:** The 'Details' tab is selected, showing the CI name 'PowerEdge T110 II' and its class 'Computer'. It also shows the 'Updated' timestamp '2026-06-12 15:57:32'.
- Key Attributes:** This section contains fields for 'Name' (PowerEdge T110 II), 'Location' (Carrera 54 No. 49 - 31, Medellin), 'Description', 'Assigned to' (Merle Wyrosdick), 'Created' (2012-02-18 00:14:43), 'Created by' (admin), and 'Updated'.
- Activity:** This section shows a list of activities related to the CI, including 'Assigned', 'Installed', 'Order received', and 'Ordered', each with a timestamp and a status.
- Service Relationships:** This section shows a diagram of the CI's relationships with other components. A message at the bottom states: 'This CI is not associated with any application services'.

5.9 Child Incident Creation

Navigate to:

Related Records → Child Incidents → New

Enter:

Field	Value
Short Description	Unable to connect to Corporate VPN
Caller	Any available user
Description	User receiving VPN authentication failure error

Save the record.

The screenshot shows the ServiceNow 'Service Operations Workspace' for an incident titled 'Unable to connect to Corporate VPN from home office'. The incident ID is INC0010003. The 'Child Incidents' section shows a table with one entry:

Number	Short description	Caller	Category	Service	Priority
INC0010007	Unable to connect to Corporate VPN	Abel Tuter	Inquiry / Help		4 - Low

The right sidebar displays 'Record Information' with SLA status (Completed/Paused) and caller details for Michael Hoefer.

5.10 Cause Documentation

From the Overview tab, select **Add Cause**.

Enter:

Probable Cause:

PowerEdge service was suspended and required restart.

Save the cause information.

5.11 Resolution Documentation

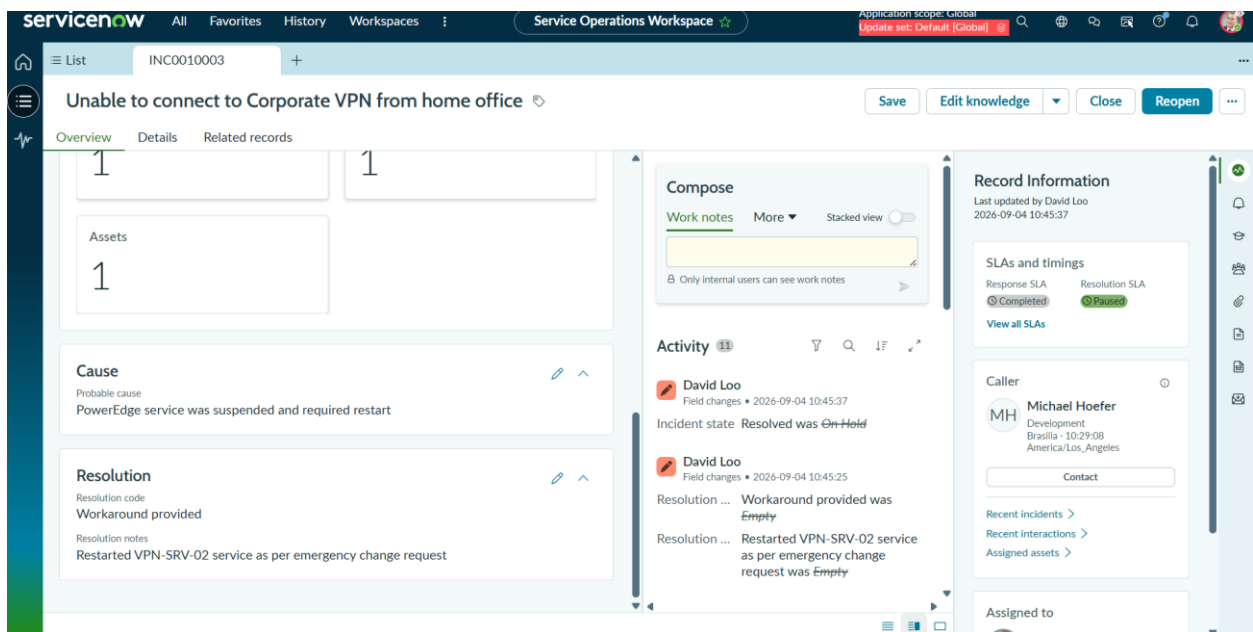
Select **Add Resolution**.

Enter:

Field	Value
Resolution Code	Workaround provided
Resolution Notes	Restarted VPN-SRV-02 service as per emergency change request.

Save the resolution.

Select **Resolve** and confirm the resolution.



5.12 Knowledge Article Creation

After resolving the incident, select **Create Knowledge**.

Configure:

Field **Value**

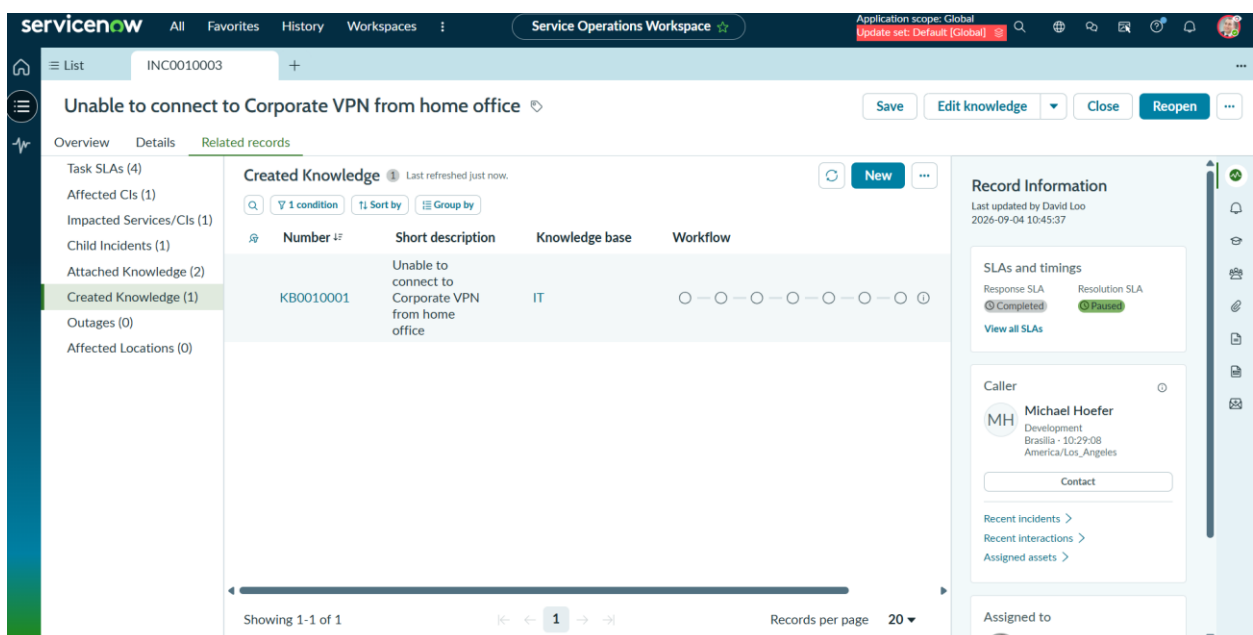
Knowledge Base IT

Article Template Standard

Select **Next**.

Verify the incident and resolution details.

Save the Knowledge Article.



5.13 Testing and Validation

Test Case	Expected Result
Incident Creation	Ticket created with caller and short description
Classification	Category, subcategory, urgency, service, offering, and CI populated correctly
Knowledge Integration	Relevant KB articles shown and attachable via Agent Assist
Reassignment	Ticket moved from Service Desk to Network
Level 2 Investigation	Full details, KB, and SLA visible to Level 2
Change Integration	Emergency Change created and linked
Child Incident	Created and linked to the parent
Resolution	Cause and resolution logged, ticket resolved
Knowledge Creation	KB article generated and linked back to incident
SLA Tracking	SLA status visible throughout via related records

Conclusion :-

The Incident Lifecycle Automation in ServiceNow successfully demonstrates a complete, structured, and traceable incident management process. By integrating Incident, Knowledge, Change, Child Incident, and SLA Management, the project helps improve resolution efficiency, documentation, and service visibility. It also supports faster and consistent incident resolution while maintaining proper documentation and a reusable knowledge base for handling similar issues in the future.